

# Multiplication Commutativity

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## 1 Introduction

This lecture describes an example of non axiomatic mathematics. We start with an inductive proof of commutativity of addition in the spirit of Peano. Then we demonstrate a non axiomatic approach. Finally we give a related example to introduce source theory as a pedagogical tool.

## 2 Example

Let addition  $+$  be defined as an associative operator. Suppose we wanted to derive commutativity.

Define  $m \times n$  as  $m$  repeated  $n$  times.

$$\begin{aligned} m \times n = & \\ & 1 + 1 + \dots + 1 + \quad (m \text{ terms}) \\ & 1 + 1 + \dots + 1 + \quad (m \text{ terms}) \\ & \vdots \\ & 1 + 1 + \dots + 1 + \quad (m \text{ terms}) \end{aligned}$$

(where the above has  $n$  rows.)

We can write the product that way because of associativity

$$(1 + 1) + 1 = (1 + 1) + 1 = 1 + 1 + 1$$

Now we prove commutativity.

First we develop a lemma that says  $m \times 1 = 1 \times m$

First we show  $m \times 1 = m$

$$\begin{aligned}
 1 = 1 + 1 + \dots + 1 &= 1 + (1 + \dots + 1) = 1 + 1 \times (m - 1) \\
 &= 2 + (1 + \dots + 1) = 2 + 1 \times (m - 2) \\
 &= k + (1 + \dots + 1) = k + 1 \times (m - k) \\
 &\vdots \\
 &= m \\
 1 \times m &= 1 + \\
 &\quad 1 + \\
 &\quad \vdots \\
 &\quad 1
 \end{aligned}$$

Using the same argument, this equals  $m$ .

Next we show  $m \times n = n \times m$  implies  $(m + 1) \times n = n \times (m + 1)$

$$\begin{aligned}
 (m + 1) \times n &= (m + 1) + (m + 1) + \dots + (m + 1) \\
 &= m + m + \dots + m + 1 + 1 + \dots + 1 && \text{(associativity)} \\
 &= m \times n + 1 \times n \\
 &= m \times n + n \\
 &= n \times m + n && \text{(from above)} \\
 &= (n + n + \dots + n) + n && \text{(m terms and 1 term)} \\
 &= n + n + \dots + n + n && \text{(m + 1 terms)} \\
 &= n \times (m + 1)
 \end{aligned}$$

Given that  $m \times 1 = 1 \times m$  and that  $m \times n = n \times m$  implies  $(m + 1) \times n = n \times (m + 1)$ , induction tells us that  $m \times n = n \times m$  for any  $m$ .

Again, given  $1 \times n = n \times 1$  and given  $m \times n = n \times m$  implies  $m \times (n + 1) = (n + 1) \times m$ , we know that  $m \times n = n \times m$  for any  $n$ .

### 3 Non axiomatic commutativity proof

Now again, we want to show  $m \times n = n \times m$ .

From the definition,

$$\begin{aligned} m \times n &= 1 + 1 + 1 + 1 \quad (m \text{ terms}) \\ &+ 1 + 1 + 1 + 1 \quad (m \text{ terms}) \\ &+ 1 + 1 + 1 + 1 \quad (m \text{ terms}) \\ &+ 1 + 1 + 1 + 1 \quad (m \text{ terms}) \\ &+ 1 + 1 + 1 + 1 \quad (m \text{ terms}) \end{aligned}$$

where there are n rows.

From the definition,

$$\begin{aligned} n \times m &= 1 + 1 + 1 + 1 + 1 \quad (n \text{ terms}) \\ &+ 1 + 1 + 1 + 1 + 1 \quad (n \text{ terms}) \\ &+ 1 + 1 + 1 + 1 + 1 \quad (n \text{ terms}) \\ &+ 1 + 1 + 1 + 1 + 1 \quad (n \text{ terms}) \end{aligned}$$

where there are m rows.

Imagine you are writing the sum on a piece of paper. Notice when you rotate the paper 90 degrees, that the sum remains the same. Therefore  $m \times n = n \times m$ .

### 4 References

[1] See Shilov - Linear Algebra (material on determinants in chapter 1 for inspiration on stepping outside of an axiomatic system to do a proof).